Pedobiologia - Journal of Soil Ecology 109 (2025) 151038

Contents lists available at ScienceDirect

Pedobiologia - Journal of Soil Ecology

journal homepage: www.elsevier.com/locate/pedobi

"Best food" does not apply to all: Quality of nematode diet for development and fitness of their predator *Parasitus* sp. (Mesostigmata: Parasitidae)

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ARTICLE INFO

Keywords:
ω3-PUFA *de novo* synthesis
Soil predatory mites
Free-living nematodes
Root-knot nematodes

ABSTRACT

Long-chain polyunsaturated fatty acids (PUFAs), especially ω3 forms, are essential nutrients for many animals and are typically obtained from dietary sources, yet there is limited availability in terrestrial ecosystem. In soil food webs, free-living nematodes are an important source of these PUFAs. Predatory mites (Mesostigmata) play a key role in regulating invertebrate populations, including free-living and plant-parasitic nematodes. Recent studies suggest that diets including nematodes improve the fitness of predatory mites and enhance their function as biological control agents within soil food webs. The predatory mite *Parasitus* sp. isolated from Mediterranean agricultural fields showed nematode consumption and colony establishment with a mixture of nematode species. This study investigated the effects of single nematode diets on the fitness of *Parasitus* sp., focusing on the roles of ω3- and ω6-PUFAs. This included one plant-parasitic and five free-living bacterial-feeding species, mostly co-occurring in the same Mediterranean fields, with one species unable to synthesize ω3-PUFAs. The results showed that nematodes rich in ω3-PUFAs supported *Parasitus* sp. development, but those high in ω6-PUFAs and deficient in ω3-PUFAs did not. However, mite survival and moulting to adulthood were generally low, possibly indicating a need for a specific stimulus or a more varied diet. This study further demonstrated ω3-PUFA *de novo* synthesis in Mesostigmata. Overall, the nematode diets influenced the development of *Parasitus* sp. differently, which emphasizes the importance of diverse nematode populations to support the fitness of higher trophic levels such as predatory mites.

1. Introduction

Polyunsaturated fatty acids (PUFAs), in particular of the omega-3 (ω3) group, are essential molecules in many animals, and play an important role in cell membrane structure, physiology and reproduction (Ruess and Müller-Navarra, 2019). For instance, in insects, PUFAs are crucial for neuronal development and reproductive success because they are precursors of signalling molecules (Malcicka et al., 2018; Stanley and Kim, 2019a). However, compared to aquatic food webs, the availability of ω3 long-chain PUFAs (ω3 LC-PUFAs) is limited in terrestrial food webs. While plants provide alpha-linolenic acid (ALA, 18:3 ω3), most animals cannot efficiently convert ALA into ω3 LC-PUFAs due to the lack of specific enzymatic pathways (Malcicka et al., 2018). Consequently, terrestrial animals obtain ω3 LC-PUFAs primarily from dietary sources, including plants and other primary producers (Buse et al., 2013; Ruess and Müller-Navarra, 2019). These essential molecules are also thought to be transferred from aquatic to terrestrial ecosystems via river predators, waterfowl and amphibian insects (Gladyshev et al., 2013).

Micro- and mesofauna might also contribute to ω3 LC-PUFAs availability in soils. Studies on different groups of invertebrates (e.g. nematodes and collembolas) have shown the presence of genes required for the *de novo* synthesis of ω3-PUFAs (Kabeya et al., 2018). Additional evidence has been observed in nematodes, mites, collembolas, gastropods, insects and earthworms, demonstrating their ability to synthesise multiple PUFAs, including in some cases eicosapentaenoic acid (EPA,

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https://doi.org/10.1016/j.pedobi.2025.151038
Received 21 November 2024; Received in revised form 18 March 2025; Accepted 19 March 2025
Available online 23 March 2025
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20:5 ω3) (Borgeson et al., 1990; Malcicka et al., 2018; Menzel et al., 2018a; Petersen and Holmstrup, 2000; Weinert et al., 1993). In particular, the bacterial-feeding nematode *Caenorhabditis elegans* (Maupas, 1899) Dougherty, 1953 (Rhabditida: Rhabditidae) possesses the complete enzymatic pathway for LC-PUFA synthesis, with similar capabilities observed in other free-living nematodes (Menzel et al., 2018a, 2019; Watts and Browse, 2002). This suggests that nematodes may play an important role in the bottom-up transfer of LC-PUFAs within soil food webs (Rueda-Ramírez et al., 2023a).

In the soil food webs, nematodes not only occupy diverse trophic roles, thus contributing to process as nutrient cycling and microbial decomposition (Ruess, 2024; Yeates, 2010), but are also an important food source for higher trophic levels, providing essential nutrients such as ω3 LC-PUFAs (Menzel et al., 2018b, 2018a; Ruess, 2024). However, their nutritional contribution to predators remains largely unexplored. Mites, particularly those belonging to Mesostigmata, Oribatida (Sarcoptiformes) and Prostigmata (Trombidiformes), are similarly integral components of soil food webs, acting as regulators of nematode populations through predation (Coleman et al., 2018; Rueda-Ramírez et al., 2023b; Walter and Ikonen, 1989; Walter and Kaplan, 1990). More than 250 species of Mesostigmata, especially Gamasina, and about 60 species of Oribatida are reported to feed on nematodes (Rueda-Ramírez et al., 2023b).

The ecological role of predatory mites as population regulators is already practically exploited in agriculture through biological control programs (Carrillo et al., 2015; Gerson et al., 2003). Studies showed that supplementing the diet of some soil predatory mite species with free-living nematodes also has positive effects on their fitness (Azevedo et al., 2020; Rueda-Ramírez et al., 2023b, 2019). Rueda-Ramírez et al. (2023a) observed that the quality of diets based on different bacterial-feeding nematodes and a plant parasite had different effects on two species of predatory mites: A diet rich in ω3-PUFAs is not necessarily the best quality food, despite the fact that these mites apparently not synthesised ω3- and ω6 LC-PUFAs. This emphasizes the importance of maintaining soil diversity to provide varied food sources for predators, matching their different nutritional needs. However, the relationship between nematode dietary composition, specifically ω3 and ω6 LC-PUFA content, and the fitness of soil predatory mites remains underexplored.

An unidentified species of *Parasitus* (Mesostigmata: Parasitidae) was found among the potential biocontrol agents in agricultural fields in the eastern Mediterranean, where previous studies identified the significance of nematode diets on predatory mites (Rueda-Ramírez et al., 2023a). Here, *Acrobeloides bodenheimeri* (Steiner, 1936) Thorne, 1937 (Rhabditida: Cephalobidae) proved to be a particularly beneficial food source. Preliminary observations indicate that this *Parasitus* sp. consumes nematodes and can establish colonies when fed a mixture of nematode species. Species of this family have an important relationship with nematodes, as immature stages seem to prefer, and in some cases require, nematodes (Castilho et al., 2015; Rueda-Ramírez et al., 2023b, 2019). Given this, further investigations were undertaken on the interactions between this predator and the nematodes co-occurring in the same agricultural fields (Rueda-Ramírez et al., 2023a) to assess whether *Parasitus* sp. exhibits similar feeding preferences and life history responses to other mite species or displays unique dietary needs.

This study focuses on nematode diet, differing in ω3 and ω6 LC-PUFA content, and the impact on the fitness of the mite *Parasitus* sp. One plant-parasitic and five free-living bacterial-feeding species were tested as prey for *Parasitus* sp., with all organisms naturally co-occurring in the same arable soil. Mite fitness was assessed based on life cycle parameters including oviposition, duration of ontological stages, stage-specific mortality and survival. The hypothesis of the study is that predatory mites reflect the ω3 LC-PUFA content of their diet, and that nematode species rich in ω3 and/or ω6 LC-PUFAs positively influence predator development and reproduction.

2. Materials and methods

2.1. Parasitus sp. stock colonies

In May 2021, specimens of *Parasitus* sp. were collected from the soil of a lettuce crop in Kfar Yehoshua (32.67353 N, 35.1520 E), Jezreel Valley, Israel. Specimens were extracted using Galen-Berlese funnels, as described in Rueda-Ramirez et al. (2022). Stock colonies were established in plastic microcosms (7 cm diameter, 4.5 cm high), whose bottom was covered with a layer of a mixture of gypsum and activated charcoal (9:1) (Abbatiello, 1965; Freire and Moraes, 2007). Bedding material (autoclaved vermiculite) was added, and colonies were maintained in total darkness at 25 ± 1°C and a relative humidity of 62 ± 10 %.

Twelve mite specimens were sent for molecular analysis to the Canadian Centre for DNA Barcoding (CCDB) (BOLD:AEZ9847, dx.doi.doi.org/10.5883/PARASI, GenBank accession number: PQ628835–PQ628845), and eleven of the same specimens (with successful DNA extraction) along with ten additional specimens were used for morphological identification. Molecular analysis revealed a 12.18 % sequence divergence (p-distance) to the closest organism, *Parasitus hyalinus* (Willmann, 1949). Morphological examination and comparison with available descriptions of species of the same genus confirmed *P. hyalinus* as the closest species, though important morphological differences were observed. Based on this evidence, we designated this species as *Parasitus* sp. for this study.

To maintain mite colonies, a mixture of all developmental stages of three nematode species, *Panagrellus nepenthicola* (Menzel, 1922) Goodey, 1945, *Panagrellus redivivus* (Linnaeus, 1767) Goodey, 1945 and *Panagrellus silusioides* Thalolikhin, 1965 (Rhabditida: Panagrolaimidae), was supplied *ad libitum* and sterile water was added daily.

2.2. Nematodes

The nematodes used in this study were the same as those used in Rueda-Ramirez et al. (2023a), which co-occur in the fields where the mites were collected. They include five species of free-living nematodes: *A. bodenheimeri*, *Oscheius tipulae* (Lam & Webster, 1971) (Rhabditida: Rhabditidae), *Pristionchus pacificus* Sommer, Letter, Kim & Sternberg, 1996 (Rhabditida: Diplogastridae) and *Panagrolaimus* sp. 1 and sp. 2 (Rhabditida: Panagrolaimidae); as well as one species of a plant-parasitic nematode, *M. incognita*. The procurement, cultivation and maintenance of nematode species are described in Rueda-Ramirez et al. (2023a).

2.3. Fatty acid extraction

Total lipid fatty acids (TLFA) were extracted from both *Parasitus* sp. and the six nematode species. Colonies and information on the total amount and pattern of TLFA of the nematodes were obtained from a previously study (Rueda-Ramírez et al., 2023a). Feeding experiments were conducted in non-choice microcosms, with mite predators and nematode prey (Ruess et al., 2004). Preliminary observations showed that *P. pacificus* caused high mite mortality, so this treatment was excluded from lipid content and fatty acid pattern analysis. Other treatments included *A. bodenheimeri* (Ab), *M. incognita* (Mi), *O. tipulae* (Ot), *Panagrolaimus* sp. 1 (P1), and *Panagrolaimus* sp. 2 (P2).

To avoid cannibalism, which occurred in groups of more than 40 individuals, cohorts of *Parasitus* sp. were reared from egg to deutonymph stage for ten days on each nematode species, with mites kept in isolation and fed *ad libitum*. Lipid content and fatty acid composition were analyzed at the deutonymph stage because moulting into adults required specific stimuli (e.g. grouping individuals of different sexes, dietary changes, or phoresy simulation), as has been observed in other Parasitidae species (Ito, 1973a, 1976, 1977a; Rueda-Ramírez et al., 2019; Yasui, 1997). Each treatment was replicated three or four times. The

number of individuals analysed per treatment and per replicate was as follows: 43, 43 and 42 individuals per replicate for Ot; 35, 33 and 30 for P1; 35, 35 and 26 for P2; 35, 34, 32 and 31 for Ab; 32, 31 and 30 for Mi.

Mites were removed from their food source two hours before analysis to allow the passage of ingested food. The mites were then weighed, freeze-dried, and stored in methanol at −20°C until lipid analysis. Stock culture mites were used as day 0 samples. Lipid extraction followed a four-step protocol (Ruess et al., 2004) using the Sherlock Microbial Identification System (MIDI). Fatty acids were methylated, extracted, and analysed using gas chromatography (GC) with flame ionization detection (FID). The identification of fatty acids was further confirmed with a mass selective detector (see Supp. Material S1). Fatty acids were classified into six groups based on chemical composition (IUPAC-IUB, 1978) as follows: *i)* Short/branched/odd fatty acids have a short, methyl-branched or odd-numbered carbon chain; *ii)* cyclic fatty acids (Cyclic) contained a cyclopropane ring; *iii)* saturated fatty acids (SFA) have no double bonds between carbon atoms; *iv)* monounsaturated fatty acids (MUFA) have one double bond in the carbon chain; and polyunsaturated fatty acids contain two or more double bonds. The latter were distinguished into *v)* ω3 polyunsaturated fatty acids (ω3-PUFA), and *vi)* ω6 polyunsaturated fatty acids (ω6-PUFA) with the terminal double bond at carbon 3 or 6, respectively, counting from the methyl end of the fatty acid.

2.4. Life cycle experiments and survival

Experiments were conducted in an incubator in darkness at 25.0 ± 0.2 °C, 70 ± 5 % RH. Life cycle experiments followed Rueda-Ramírez et al. (2019), starting with a synchronisation of the mite cohorts (OECD, 2016). For this, gravid females were isolated in small arenas (3.5 cm diameter, 1 cm high), the bottom of which was covered with a layer as described for the rearing microcosms. The females were separated into six groups, according to the number of treatments (the same four free-living nematode species and *M. incognita* used for the determination of lipid content and fatty acid pattern + *P. pacificus* (Pp)). Each of the nematodes were supplied to each group of females as food and sterile water was added daily. After 24 h, females were returned to stock colonies.

Initially, the eggs laid by the females were kept in the synchronization arenas due to a high mortality during the transfer to other units. We therefore left the eggs in the original unit until they hatched into larvae. More nematodes from each treatment were added to the synchronization arenas to ensure food when the larvae hatched. After 24 h, larvae were isolated in similar arenas. The experiment started with the following number of larvae for each treatment: 66 for Ab, 96 for Mi, 52 for Ot, 64 for P1, 67 for P2 and 65 for Pp. Nematodes from the corresponding treatment were supplied *ad libitum* daily. At the same time, ontogenetic stages and mortality were recorded daily using a stereomicroscope (up to 50x) based on morphology and exuviae. A mite was considered dead when unresponsive after stimulation with a fine brush and the legs were curved downwards. For each stage, the duration of the stage and mortality were determined.

At the deutonymph stage, mites were paired with individuals of the opposite sex from the same treatment to promote adult development (based on Rueda-Ramirez et al. (2019), and personal observations). This approach was based on the confirmation that a stimulus (e.g. grouping of individuals of different sexes, change of diet or simulation of phoresy) is required for the deutonymphs to molt into adults, as has been observed in other Parasitidae species (Ito, 1973a, 1976, 1977a; Rueda-Ramírez et al., 2019; Yasui, 1997). Adults were fed and examined daily to record longevity, survival, sex ratio and oviposition. Individual monitoring was possible even when mites were paired, as their sex could be determined from the deutonymph stage onwards. Females were paired with males from the same treatment or, if unavailable, with stock males. Eggs were counted daily.

Some mites disappeared during the experiment and were treated as censored data. Only mites that visibly reached the next stage were used for stage duration analysis. Total immature stage duration was based on mites that fully developed from egg to adult. Censored data were excluded from the determination of stage duration and no life table parameters were calculated.

2.5. Statistical analysis

Fatty acids: After obtaining the lipid patterns of the mites for each treatment, all fatty acids that were not present in at least one of the samples with a frequency greater than 1 % were discarded from further analysis. This resulted in a lipid library of 27 fatty acids for *Parasitus* sp. feeding on the five different nematode species, represented by 16 samples (three replicates for each nematode diet, except in the case of Ab with four replicates).

The means of the different variables (TLFA, biomass, content of each fatty acid, proportion of each fatty acid, content of each of the groups/fractions of the fatty acids, proportion of each of the groups/fractions of the fatty acids, difference of the proportion of each of the groups/fractions of the fatty acids between predator and prey) were compared between treatments (nematode diet) using Generalized Linear Models (GLM) fitted with different distributions (Gaussian, Gamma, Quasibinomial; indicated in Supplementary material or in results) and nematode species diet (treatment) as a factor. Different distributions were used to ensure that the assumptions of each model were met, leading to more accurate and interpretable results. When significant differences at a $p \leq 0.05$ level were detected, a Tukey multiple comparison test was performed.

Statistical analyses were performed using the software R (version 4.2.2, 2022) with packages gamlss, ggplot2, lme4, car, MASS, multcomp, ExpDes, vegan, and hnp. The variation is given as the standard error of the mean (SE).

Developmental stages and mortality: The mean duration of each stage and of the entire immature phase (larva–adult) as well as the longevity of adults were compared between treatments using a Kruskal-Wallis test, followed by a Dwass-Steel-Critchlow-Fligner pairwise comparison in the case of significant differences.

To compare the mortality of each developmental stage between treatments, the total number of mites that died at each stage for each treatment was recorded and presented in a contingency table. A χ^2 test was performed to test for significant differences in stage-specific mortality, and where significance was found, two by two tables were used to compare the mortality of each stage for each treatment with the total mortality of all other stages in all other treatments. A further χ^2 test was performed with these smaller contingency tables. In addition, a Kaplan-Meier survival curve was constructed for each treatment group with data from larval to adult or last live stage. The log-rank test was used to compare the survival curves for each treatment. The results were statistically significant at a level $p \leq 0.05$.

Jamovi statistical software was used for these analyses (based on R Core Team, 2021; The Jamovi Project, 2021).

3. Results

3.1. Lipid content and fatty acid pattern in Parasitus sp

The nematode diet had significant effect on the TLFA content (Gamma dist., $\chi^2_{4,16} = 138.13$, $p < 2.2 \text{ X } 10^{-16}$), but not on the dry weight per individual mite (biomass) (Gaussian dist. $F_{4,16} = 2.35$, $p = 0.14$). Mites fed with *Panagrolaimus* sp. 1 and *Panagrolaimus* sp. 2 had the highest lipid content, which was significantly larger compared to the mites fed with *M. incognita* and *A. bodenheimeri*. *Parasitus* sp. fed with *O. tipulae* had an intermediate lipid content (Table 1).

The mites largely reflected the fatty acid pattern of the nematode diet, being most prominent with *M. incognita* as food source (Fig. 1). The relative amount of short, branched, or odd-numbered fatty acids was

Table 1
Mean dry weight (μg ± SE) and content (nmol mg−1 DW ± SE) of total lipid fatty acid (TLFA) of the predatory mite, *Parasitus* sp. feeding on five different nematode species: *Meloidogyne incognita* (Mi), *Acrobeloides bodenheimeri* (Ab), *Oscheius tipulae* (Ot), *Panagrolaimus* sp. 1 (P1), and *Panagrolaimus* sp. 2 (P2).

	Root-knot nematode	Free-living nematodes			
	Mi	**Ab**	**Ot**	**P2**	**P1**
Biomass/mite (μg dry weight)	4.57 ± 0.35 a	5.25 ± 0.09 a	4.76 ± 0.24 a	5.4 ± 0.04 a	3.99 ± 0.46 a
TLFA (nmol mg^{-1} DW)	10.76 ± 0.83 b	14.84 ± 1.77 b	60.95 ± 7.56 a	101.55 ± 28.43 a	94.5 ± 16.99 a

Letters following the mean values indicate significant differences between nematode diets for each variable (row) (Tukey's post-hoc comparisons test after GLM with different distributions, $p < 0.05$).

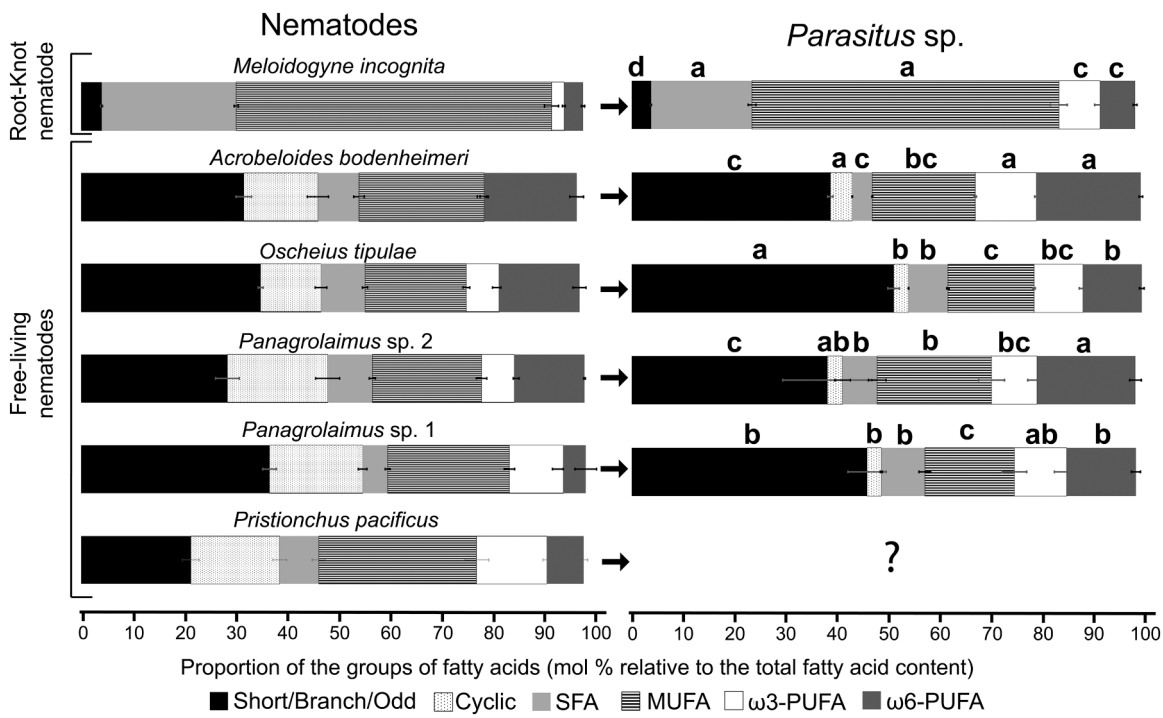

Fig. 1. Proportion (%) of total lipid fatty acids (TLFA) of predatory mite *Parasitus* sp. (right) and of the nematode diet (left). Short/branch/odd fatty acids - 10:0 3OH, 12:0, **14:0**, **14:0 anteiso**, 14:0 iso, 14:0, 14:0–3OH, 15:0 anteiso, 15:0 iso, **16:0 anteiso**, 16:0 iso, 17:0 anteiso, 17:0 iso, 17:0, **18:0 iso**, 19:0 iso; Cyclic fatty acids - 17:0 cyclo ω7, 19:0 cyclo ω7, 20:0 cyclo ω6; SFA - 16:0, 18:0, 20:0, 22:0; MUFA - 14:1 ω7, **16:1 ω5**, 16:1 ω7, **17:1 ω8**, 18:1 ω5, 18:1 ω7, 18:1 ω9, 20:1 ω6, 20:1 ω7, 22:1 ω6, 22:1 ω9; ω3-PUFA – **18:3 ω3**, 20:4 ω3, 20:5 ω3; ω6-PUFA - 18:2 ω6, 18:3 ω6, 20:2 ω6, 20:3 ω6, 20:4 ω6, **21:3 ω6**. Fatty acids highlighted only in bold correspond to those present only in *Parasitus* sp.; fatty acids underlined correspond to those present only in the nematode diet. Different letters in each fatty acid group between diets indicate significant differences (GLM followed by Tukey's with $p < 0,05$). Data on the fatty acid pattern and proportion of fatty acid groups of nematode diets obtained from Rueda-Ramírez et al. (2023a). The question mark (?) indicates that the fatty acid content is unknown because it could not be determinated due to high mortality.

lowest in mites fed on *M. incognita* (about 4 %), and highest in those fed on *O. tipulae* (about 50 %), followed by mites feeding on *Panagrolaimus* sp. 1 (about 46 %). In mites reared on *M. incognita*, no cyclic fatty acids were detected, which were was also not observed in the nematode. In mites fed on the other nematodes, the proportion of cyclic fatty acids was less than 5 %. The relative amount of SFAs was significantly higher in mites feeding *M. incognita* (almost 20 %), and significantly lower in those consuming *A. bodenheimeri* (about 4 %). Relative MUFA contents were highest in mites with *M. incognita* (about 60 %) as diet, followed by mites fed on *Panagrolaimus* sp. 2 (about 22 %) and *A. bodenheimeri* (about 20 %). The highest relative content of ω3-PUFAs was detected in mites reared with *Panagrolaimus* sp. 1 (about 10 %) and, *A. bodenheimeri* (about 12 %), although the latter lacks these fatty acids, and was lowest in those fed on *M. incognita* (about 8 %). Omega-6-PUFA relative contents were found to be highest in mites fed on *A. bodenheimeri* (about 20 %) and *Panagrolaimus* sp. 2 (about 19 %), as observed in the nematodes, and lowest in mites fed on *M. incognita* (about 7 %) (Fig. 1, Supp. Material S2.1 and S3.1).

Stearic acid (18:0) and vaccenic acid (18:1 ω7) were detected in high quantities in mites that fed on *M. incognita* (14 %,1.6 ± 0.3; and 47 %, 5.2 ± 0.7 nmol mg^{-1} DW, respectively), coinciding with the high amount observed in the nematodes. Oleic acid (18:1 ω9) was detected in low amounts in mites feeding *M. incognita* (4 %, 0.5 ± 0.1 nmol mg^{-1} DW), and in high amounts in mites feeding *Panagrolaimus* sp. 1 (12 %, 11.7 ± 4.3 nmol mg^{-1} DW) and *Panagrolaimus* sp. 2 (12 %, 12.9 ± 7.7 nmol mg^{-1} DW) as diet (Supp. Material S2.2).

The fatty acid eicosatetraenoic acid (ETA, 20:4 ω3) was detected in high relative amounts in mites fed on *A. bodenheimeri* (9 %, 1.3 ± 0.3 nmol mg^{-1} DW), and in lower amounts in those fed on *Panagrolaimus* sp. 1 (4 %, 3.6 ± 1.2 nmol mg^{-1} DW) and *M. incognita* (1 %, 0.2 ± 0.1 mmol/g DW) (Gaussian distr., $F_{4,16} = 129.97$, $p = 3.51 \times 10^{-09}$). ALA (18:3 ω3) was detected in higher relative amounts in mites feeding *A. bodenheimeri* (3 %, 0.5 ± 0.1 nmol mg^{-1} DW), whereas it was not detected with *M. incognita* as diet (Gamma distr., $\chi^2_{3,13} = 23.96$, $p = 2.54 \times 10^{-5}$). In contrast, EPA (20:5 ω3) was absent in mites reared with *A. bodenheimeri* and rare in those fed on *Panagrolaimus* sp. 2 (<1 %, 0.2 ± 0.2 nmol mg^{-1} DW), but occurred in highest relative amounts in mites fed on *M. incognita* (7 %, 0.7 ± 0.1 nmol mg^{-1} DW), followed by

Panagrolaimus sp. 1 (4 %, 3.96 ± 0.82 nmol mg^{-1} DW) (Gamma distr., $\chi^2_{4,16} = 435.1$, $p < 2.2 \times 10^{-16}$). No fatty acid 20:3 ω3 was detected in any of the mites (Supp. Material S2.2, Fig. 2a).

The fatty acid LA (18:2 ω6) was observed in highest relative amounts in mites fed on *A. bodenheimeri* (13 %, 2.0 ± 0.5 nmol mg^{-1} DW) and *Panagrolaimus* sp. 2 (10 %, 11.4 ± 6.5 nmol mg^{-1} DW), and in low amounts in those on *M. incognita* (4 %, 0.4 ± 0.1 nmol mg^{-1} DW) (Gaussian distr., $\chi^2_{4,16} = 304.3$, $p < 2.2 \times 10^{-16}$). Gamma-Linolenic Acid (GLA,18:3 ω6) was detected in higher relative amounts with *Panagrolaimus* sp. 2 (4 %, 4.2 ± 2.3 nmol mg^{-1} DW) and *O. tipulae* (3 %, 2.0 ± 0.2 nmol mg^{-1} DW) compared to *M. incognita* (<1 %, 0.1 ± 0.02 nmol mg^{-1} DW) as diet (Gaussian distr., $\chi^2_{4,16} = 129.01$, $p < 2.2 \times 10^{-16}$). Arachidonic Acid (ARA, 20:4 ω6) occurred in considerable higher proportions in mites fed on *Panagrolaimus* sp. 2 (5 %, 5.0 ± 3.2 nmol mg^{-1} DW) and *A. bodenheimeri* (5 %, 0.8 ± 0.2 nmol mg^{-1} DW), but was absent when cultured on *O. tipulae* (Gamma distr., $\chi^2_{4,16} = 15.77$, $p = 0.0013$) (Supp. Material S2.2, Fig. 2b).

3.2. Differences in fatty acid patterns between nematodes and Parasitus sp

The proportion and composition of all short, branched, or odd-numbered fatty acids differed between *Parasitus* sp. and the bacterial-feeding free-living nematodes, but not for the root-knot nematode *M. incognita*. There was also a significant increase in the overall proportion of these fatty acids in all mites that fed on free-living nematodes compared to the nematodes themselves (Gaussian distr., $\chi^2_{1,16} = 6.6 - 42.9$, $p = 9.95 \times 10^{-7} - 0.01$), which was most pronounced in mites fed on *O. tipulae* (Gamma distr., $\chi^2_{4,16} = 96.38$, $p < 2.2 \times 10^{-16}$) (Table 2).

The relative amount of cyclic fatty acids decreased in all mites compared to the proportion present in the nematodes diet, with the greatest reduction with *Panagrolaimus* sp. 1 and *Panagrolaimus* sp. 2 as diet (Quasibinomial distr., $\chi^2_{4,16} = 915.47$, $p < 2.2 \times 10^{-16}$). Also, the proportion of SFAs was reduced in all mites relative to nematodes, except when feeding on *Panagrolaimus* sp. 1, where SFA increased in proportion. The greatest decline in SFAs was observed for mites fed on *A. bodenheimeri* and *M. incognita* (Gaussian distr., $\chi^2_{4,16} = 122.49$, $p < 2.2 \times 10^{-16}$) (Table 2). Of these fatty acids, the amount of stearic acid (18:0) was lower in mites fed on free-living nematodes compared to their prey, but a higher amount of stearic acid was observed in mites fed on *M. incognita* compared to the nematode itself.

As with the SFAs, the proportion of MUFAs was reduced for all mites compared to the nematodes, except when feeding *Panagrolaimus* sp. 2, with an increase in mites of 1 % relative to nematodes. The greatest decrease in mites was observed when feeding on *A. bodenheimeri* and *O. tipulae* (Gaussian distr., $\chi^2_{4,16} = 39.76$, $p = 4.85 \times 10^{-8}$) (Table 2).

Table 2
Difference in the proportion (% ± SE) of the groups/fractions of fatty acids found in *Parasitus* sp. feeding on each of five different species of nematodes in relation to those reported in the nematodes themselves: *Meloidogyne incognita* (Mi), *Acrobeloides bodenheimeri* (Ab), *Oscheius tipulae* (Ot), *Panagrolaimus* sp. 1 (P1), and *Panagrolaimus* sp. 2 (P2).

	Root-knot nematode	Free-living nematodes			
	Mi	Ab	Ot	P2	P1
Short/ branch/ odd	−0.3 ± 0.1c	7.2 ± 0.2 b	16.3 ± 0.2 a	7.3 ± 1.6 b	9.4 ± 2.3 b
Cyclic	0 a	−10.1 ± 0.05 b	−8.8 ± 0.05 b	−16.5 ± 1.5c	−15.3 ± 0.1c
SFA	−6.4 ± 0.5 d	−4.1 ± 0.1 cd	−0.9 ± 0.2 b	−1.9 ± 1.3 bc	3.6 ± 0.8 a
MUFA	−1.3 ± 0.8 ab	−4.2 ± 0.2 bc	−2.9 ± 0.2 bc	1.1 ± 1.5 a	−6.2 ± 1.2c
ω3-PUFA	5.6 ± 0.4 b	12 ± 0.3 a	3.2 ± 0.2c	2.5 ± 0.5c	−0.2 ± 0.5 d
ω6-PUFA	3.2 ± 0.3 b	2.5 ± 0.1 b	−4.1 ± 0.7c	5.6 ± 1.4 ab	5.6 ± 1.4 a

Letters following the mean values indicate significant differences between nematode diets for each variable (row) (Tukey's post-hoc comparisons test after GLM with different distributions [Gamma, Gaussian, Quasibinomial], $p < 0.05$). SFA: saturated fatty acids, MUFA: monounsaturated fatty acids, ω3-PUFA: omega 3 polyunsaturated fatty acids, ω3-PUFA: omega 6 polyunsaturated fatty acids.

The proportion of ω3-PUFAs was higher in mites fed on *A. bodenheimeri*, followed by *O. tipulae*, compared to the nematodes themselves (>6 %). Mites fed on *Panagrolaimus* sp. 2 and *M. incognita* increased the proportion of these fatty acids by 3 % compared to their nematode diet, while no change was observed with *Panagrolaimus* sp. 1 (Gaussian distr., $\chi^2_{4,16} = 605.68$, $p < 2.2 \times 10^{-16}$) (Table 2). The large increase in mites fed on *A. bodenheimeri* was due to the presence of the fatty acids ALA and ETA, which were absent in the nematodes, as well as other ω3-PUFAs. The fatty acid ALA was not detected in any of the nematode species but was detected in all mites feeding on free-living nematodes. The relative amount of EPA was lower in all mites that fed on free-living nematodes containing ETA, but the decrease was not as strong in mites feeding on *Panagrolaimus* sp. 1 (decrease of only 0.69 %). A decrease in the relative amounts of ETA was only observed in mites fed on this nematode (decrease of 1.77 %) (Fig. 3).

The proportion of ω6-PUFA increased in all mites relative to the proportion found in the nematodes, except for mites fed on *O. tipulae* with 4 % less compared to their diet. These higher proportions were significant for mites feeding on *Panagrolaimus* sp. 1 and *Panagrolaimus*

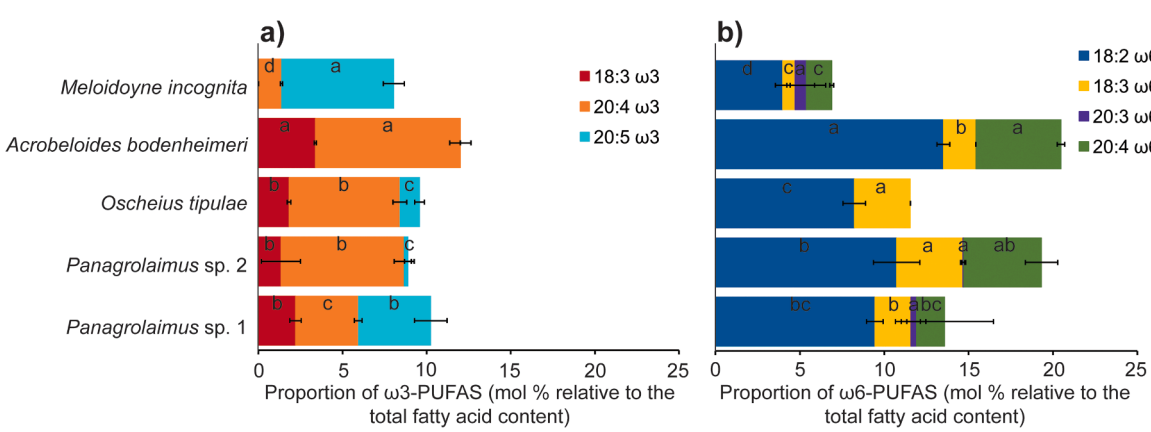

Fig. 2. Proportion (%) of **a)** ω3-Polyunsaturated fatty acids (PUFAS) and **b)** ω6-PUFAS of deutonymphs of predatory mite *Parasitus* sp. fed on each of the diet nematode species. Letters on each bar indicate significant differences between nematode diets for each fatty acid (Tukey's post-hoc comparisons test after GLM with different distributions [Gamma, Gaussian], $p < 0.05$).

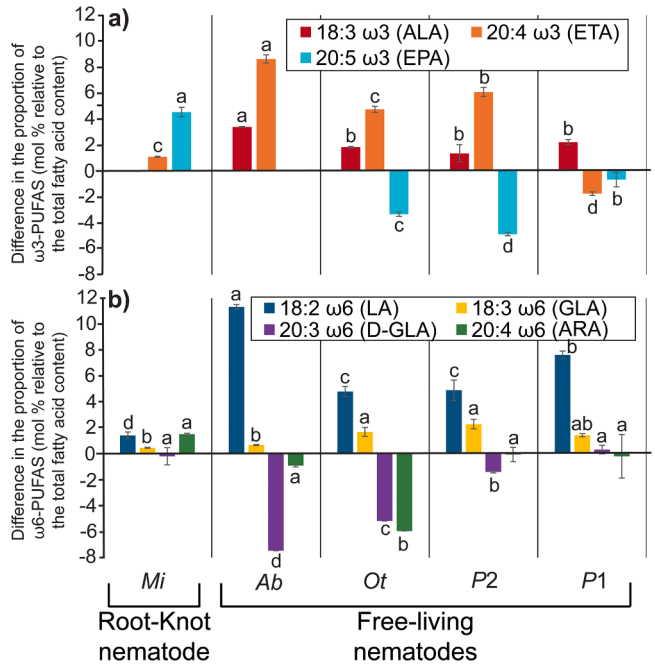

Fig. 3. Differences in the proportion of **a)** ω3-Polyunsaturated fatty acids (ω3-PUFA) and **b)** ω6-Polyunsaturated fatty acids (ω6-PUFA) between the mite *Parasitus* sp. and its nematode prey. The root-knot nematode: *Meloidogyne incognita* (Mi); and the free-living nematodes: *Acrobeloides bodenheimeri* (Ab), *Oscheius tipulae* (Ot), *Panagrolaimus* sp. 1 (P1), *Panagrolaimus* sp. 2 (P2), and *Pristionchus pacificus* (Pp). A positive difference means a higher proportion of a fatty acid in the mite compared to its prey, a negative change a lower proportion of a fatty acid in the mite. Letters on each bar indicate significant differences between nematode diets for each fatty acid (Tukey's post-hoc comparisons test after GLM with different distributions [Gamma, Gaussian], $p < 0.05$).

sp. 2 (Gaussian distr., $\chi^2_{4.16} = 79.37$, $p = 2.4 \times 10^{-16}$) (Table 2). A common feature across all treatments was an increase in the relative amount of LA (18:2 ω6) in *Parasitus* sp. However, this increase was less pronounced in mites fed on *M. incognita* compared to those fed on free-living nematodes. The fatty acid GLA (18:3 ω6) was found in higher proportions in all mites compared to their prey, especially when feeding on *O. tipulae*, *Panagrolaimus* sp. 1, and *Panagrolaimus* sp. 2. Dihomo-γ-linolenic acid (D-GLA, 20:3 ω6) was detected in mites fed on *Panagrolaimus* sp. 1, although it was absent in these nematodes. In contrast, this fatty acid showed the greatest reduction relative to the diet in mites fed *A. bodenheimeri* (Fig. 3).

3.3. Biological parameters of the consumer predatory mite *Parasitus* sp

The different nematode diets affected the postembryonic developmental time, survival and mortality of *Parasitus* sp. The postembryonic development time was significantly longer when mites were fed on *M. incognita* ($\chi^2_{4,71} = 28.84$, $p < 0.001$), due to extended durations of the larval ($\chi^2_{5325} = 34.5$, $p < 0.001$), protonymphal ($\chi^2_{5237} = 81.4$, $p < 0.001$), and deutonymphal ($\chi^2_{4,71} = 23.76$, $p < 0.001$) stages. Mites feeding on *A. bodenheimeri* also exhibited the second-longest post-embryonic stage duration, followed by those feeding on *O. tipulae*. The total duration of this immature period was significantly shorter when mites consumed *Panagrolaimus* sp. 1, followed by *Panagrolaimus* sp. 2 (Table 3).

Most of the nematode species, when offered individually to the mites, were sufficient to support development to the adult stage. However, among the adults observed, the longevity of the adult stage was shorter when mites were reared on *Panagrolaimus* sp. 1 and *Panagrolaimus* sp. 2, and longer with *A. bodenheimeri* ($\chi^2_{4,62} = 16.14$, $p = 0.003$). No significant differences were observed in adult longevity when mites were fed *O. tipulae* and *M. incognita*. Regarding male longevity, no significant differences were found with *O. tipulae*, *Panagrolaimus* sp. 1, and *M. incognita* as diet. However, male longevity increased when mites fed on *A. bodenheimeri* but decreased with *Panagrolaimus* sp. 2 ($\chi^2_{4,41} = 11.84$, $p = 0.019$). No significant differences were observed in the longevity of adult females ($\chi^2_{4,21} = 7.97$,

Table 3
Mean duration (days ± SE) and mortality (% relative to the initial N of the stage) of the stages, sex ratio, oviposition and fecundity of the predatory mite, *Parasitus* sp. feeding on six different nematode species: *Meloidogyne incognita* (Mi), *Acrobeloides bodenheimeri* (Ab), *Oscheius tipulae* (Ot), *Panagrolaimus* sp. 1 (P1), *Panagrolaimus* sp. 2 (P2), and *Pristionchus pacificus* (Pp).

		Root-knot nematode	Free-living nematodes				
		Mi	*Ab*	*Ot*	*P2*	*P1*	*Pp*
Larval stage	Duration [days]* (N)	1.57 ± 0.07 a (67)	1.11 ± 0.05c (53)	1.39 ± 0.08 ab (44)	1.38 ± 0.07 ab (55)	1.17 ± 0.05 bc (58)	1.38 ± 0.08 ab (48)
	% mortality[ǂ]	35.1 a	15.2 b	2.9c	5.1 b	5.0 b	22.6 b
Protonymphal stage	Duration [days]* (N)	3.00 ± 0.10 a (44)	2.28 ± 0.11 bc (39)	1.89 ± 0.13 cd (37)	2.61 ± 0.11 b (46)	1.73 ± 0.08 d (55)	2.13 ± 0.21 bd (16)
	% mortality[ǂ]	21.6 b	0.0 d	17.1c	17.9c	5.0 b	50.9 a
Deutonymphal stage	Duration [days]* (N)	13.3 ± 5.78 ab (3)	14.6 ± 1.03 a (10)	8.09 ± 2.10 ab (11)	7.24 ± 0.52 b (17)	5.60 ± 0.57 b (30)	- (0)
	% mortality[ǂ]	40.5 b	54.5 a	45.7 b	35.9 b	15.0c	26.4 b
Preadult stage	Duration [days]* (N)	19.0 ± 5.51 a (3)	18.8 ± 1.07 ab (10)	12.6 ± 1.99 abc (11)	12.2 ± 0.49 bc (17)	9.50 ± 0.56c (30)	- (0)
	% mortality[ǂ]	97.3	69.7	65.7	59	25	100
Adult stage (female and male) longevity [days]* (N)		26.3 ± 10.2 ab (3)	26.8 ± 1.88 a (9)	17.6 ± 2.15 ab (9)	16.6 ± 0.75 b (15)	18.0 ± 1.01 b (26)	-
Female lifespan [days]* (N)		46.0 a (1)	28.3 ± 5.04 a (3)	15.3 ± 4.84 a (3)	17.8 ± 1.70 a (4)	16.9 ± 1.49 a (10)	- (0)
Male lifespan [days]* (N)		16.5 ± 4.50 ab (2)	26.0 ± 1.75 a (6)	18.7 ± 2.39 ab (6)	16.2 ± 0.83 b (11)	18.7 ± 1.34 ab (16)	- (0)
Sex ratio (female:male)		1:2	3:7	3:8	4:13	4:6	-
Oviposition period [days]				3.00 ± 0.58		6.10 ± 0.98	
Fecundity [eggs/female/day]				0.17 ± 0.17		0.95 ± 0.39	

Letters following the mean values indicate significant differences between nematode diets for each variable (row) (*Dwass-Steel-Critchlow-Fligner pairwise comparison after Kruskal-Wallis test, $p < 0.05$; [ǂ]χ^2-test of association for contingency tables, $p < 0.05$.)

$p = 0.093$) (Table 3).

A significant difference in mortality across all stages was observed in all treatments (χ^2 test of association: N = 237; $\chi^2 = 122$, $p < 0.001$). Mortality during the different postembryonic stages was highest in mites feeding on *M. incognita*, *A. bodenheimeri*, and *P. pacificus* (Table 3). The highest mortality in the larval stage occurred when mites were fed on *M. incognita*, while the highest mortality in the protonymphal stage was observed in mites fed on *P. pacificus*. With the latter, none of the mites survived beyond the protonymph stage. At the deutonymphal stage, mites fed on *A. bodenheimeri* had the highest mortality. Mites fed *Panagrolaimus* sp. 1 exhibited the lowest mortality across all stages except the larval stage (Table 3).

Significant differences were observed in the survival curves of *Parasitus* sp. across treatments (χ^2= 136, d.f. = 5, $p < 0.001$). The survival curve was shorter when mites were fed on *P. pacificus* and *Panagrolaimus* sp. 2, and longer when fed on *M. incognita* ($p < 0.05$). Despite the high mortality rates during the immature stages of mites fed on *M. incognita*, their survival curve was prolonged, primarily due to the extended lifespan of a single female and the prolonged duration of the preadult stage (larva–deutonymph), particularly the deutonymphal stage in some mites. In mites fed on *Panagrolaimus* sp. 2, although mortality was lower than in those fed on *A. bodenheimeri* and *O. tipulae*, a significantly distinct survival curve was observed. This was mainly attributed to the short duration of the adult stage and overall short lifespan (Table 3, Supp. Material S3.2). No other significant differences in survival were observed among mites reared on *O. tipulae*, *A. bodenheimeri*, and *Panagrolaimus* sp. 1 (Fig. 4).

A male-biased sex ratio was observed in all treatments, except for *P. pacificus*, where none of the mites reached the adult stage. Oviposition was only observed in females fed *O. tipulae* and *Panagrolaimus* sp. 1, with fecundity being less than one egg per female per day (Table 3).

4. Discussion

4.1. The lipid profile of Parasitus sp. deutonymphs reflects that of their prey

Corresponding to a previous study investigating other Mesostigmata species (Rueda-Ramírez et al., 2023a), a clear link between the lipid pattern of mites and their nematode prey was observed. The fatty acid pattern of *Parasitus* sp. deutonymphs reflected the trophic origin of their prey, distinguishing between nematodes from the detritivore or herbivore food chains. For instance, the lipid composition of *Parasitus* sp. feeding on *M. incognita* closely mirrored that of their prey, especially in MUFAs and SFAs. Additionally, the overall proportion of fatty acids indicative of bacteria (e.g. iso, anteiso, or cyclic molecules; Moore and Floss, 1999; Zelles, 1999) was very low in both *M. incognita* and mites reared on this nematode (see Supp. Material S3.2). This is likely due to the nutrition via plant tissue of *M. incognita,* in which these fatty acids are generally not present (Ruess and Chamberlain, 2010). Thus, the lipid pattern of the free second stage juveniles still reflects the diet of the root-feeding parent generation. In contrast, the fatty acid composition of the bacterial-feeding nematodes was only partially mirrored by their predator, suggesting *de novo* synthesis of certain fatty acids by the mites for specific metabolic needs. In the oribatid mite *Archegozetes longisetosus* Aoki, 1965 (Sarcoptiformes: Oribatida: Trhypochthoniidae), the nonpolar lipid fatty acid (NLFA) pattern has been shown to reflect the feeding guild, yet this was tested on a wide range of resources of animal, bacterial, fungal and herbal origin (Brückner et al., 2017; Brückner and Heethoff, 2020).

4.2. MUFA and SFA composition could be indicators of feeding guild

In particular, MUFAs showed distinct trends depending on the nematode prey. For instance, mites feeding on *O. tipulae* and two *Panagrolaimus* species exhibited an increase in oleic acid (18:1 ω9), while mites preying on *A. bodenheimeri* showed a decrease. This points to a specific metabolic role of oleic acid, e.g. it may serve as a precursor for PUFA biosynthesis, as reported for Collembola (Menzel et al., 2017). Moreover, mites fed on *M. incognita* displayed high proportions of MUFAs, particularly 18:1 ω7, which is dominant in this nematode species (Krusberg et al., 1973; Sekora et al., 2010, 2009). This suggests that the ratio of MUFAs to other fatty acids could serve as an indicator of trophic relationships, distinguishing predators in the herbivore versus detritivore food chains. However, the present work and that of Rueda-Ramírez et al. (2023a) together investigated this trophic link only for one plant-feeding nematode and three mite predatory species, and further studies are needed to assess the quality of such a MUFA indicator.

The proportion of SFAs also varied with prey type. For example, mites feeding on *A. bodenheimeri* showed reduced stearic acid content compared to their nematode prey. This reduction may result from stearic acid being metabolized into oleic acid and subsequently converted into essential PUFAs such as LA and ALA (Malcicka et al., 2018; Ruess and Chamberlain, 2010). In mites feeding on *M. incognita*, SFA levels remained high, mirroring the composition of the nematode. This again highlights the dietary routing of fatty acids from prey to predators, which can help to elucidate the trophic relationships of mites. However,

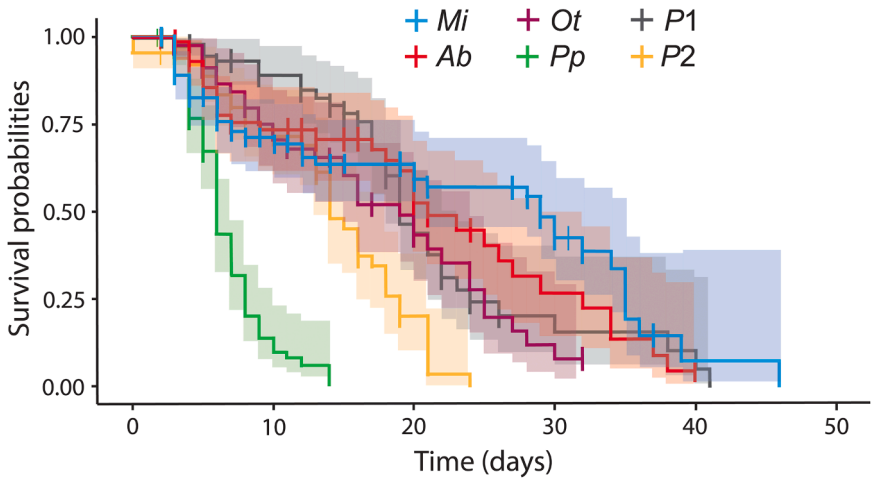

Fig. 4. Kaplan-Meier survival curve for *Parasitus* sp. from larva to adult fed on the root-knot nematode: *Meloidogyne incognita* (Mi); and the free-living nematodes: *Acrobeloides bodenheimeri* (Ab), *Oscheius tipulae* (Ot), *Panagrolaimus* sp. 1 (P1), *Panagrolaimus* sp. 2 (P2), and *Pristionchus pacificus* (Pp). The confidence intervals are represented as the translucid area around each curve.

unique fatty acids in *Parasitus* sp. deutonymphs, that are absent in their prey, suggest *de novo* biosynthesis (Iverson et al., 2004; Schulz, 2008) or potential interactions with mite-associated microbiota. Although numerous microorganisms have been identified in association with Mesostigmata mites (Chaisiri et al., 2015; Merlin et al., 2022; Perotti and Braig, 2011, 2004; Sumner-Kalkun et al., 2020), their role in shaping the lipid profiles of predatory soil mites remains to be explored.

4.3. De novo synthesis of ω3 and ω6 LC-PUFAs in Parasitus sp

De novo biosynthesis of PUFAs, including ω3 LC-PUFAs, was particularly evident in *Parasitus* sp. deutonymphs fed on free-living bacterial-feeding nematodes, as the absence of ALA in this diet, can reveal its synthesis by the mites. Likewise, *de novo* synthesis was especially pronounced in mites fed with *A. bodenheimeri*, in which ω3 LC-PUFA were not present, coinciding with low proportions of their metabolic precursors, oleic and stearic acids (Malcicka et al., 2018). The reduction of D-GLA (20:3 ω6) in mites fed on *A. bodenheimeri*, accompanied by an increase of ETA relative to the diet, may also indicate the presence of desaturases (Kabeya et al., 2018; Monroig et al., 2022). In mites fed with the plant parasite *M. incognita*, the increase of EPA in the proportion relative to the diet may be due to the low total lipid content (TLFA) of the mites. Nevertheless, apparent ARA synthesis was still observed in *Parasitus* sp.

The ability to synthesise essential PUFAs *de novo* has been reported in several invertebrate species. Cripps et al. (1986) and Borgeson et al. (1990) reported LA synthesis in several insect species, including *Periplaneta americana* (Linnaeus, 1758) (Insecta: Blattodea). Similarly, Blaul and Ruther (2011) identified LA synthesis in *Nasonia vitripennis* (Walter, 1836) (Insecta: Hymenoptera). Additionally, PUFA synthesis has been observed in Collembola (Malcicka et al., 2017; Menzel et al., 2018a). The nematode *C. elegans* can synthesise a wide range of PUFAs *de novo*, with genetic evidence suggesting that other free-living nematodes may also have the capacity for ω3 LC-PUFA synthesis (Menzel et al., 2018a; Watts and Browse, 2002). However, PUFA metabolism in arachnids has been less extensively studied. Some spiders, including members of the families Araneidae, Tetragnathidae and Linyphiidae (Arachnida: Araneae), have been observed to produce EPA from dietary precursors (Mathieu-Resuge et al., 2022). This study provides further evidence for the synthesis of these important molecules in arachnids.

In mites, we recently suggested the absence of ω3 desaturases (FAT-1) due to evidence of the synthesis of ω6-PUFAs, but no evidence of the ability to synthesize ω3-PUFAs (Rueda-Ramírez et al., 2023a). For example, the Astigmatina species *Tyrophagus similis* Volgin, 1949, *Tyrophagus putrescentiae* (Schrank, 1781) and *Carpoglyphus lactis* (Linnaeus, 1767) (Acari: Sarcoptiformes: Oribatida: Acaridae) have been observed to be able to synthesise LA *de novo* (Aboshi et al., 2013; Shimizu et al., 2014), but no ω3-PUFA synthesis has been reported. Likewise, the oribatid mite *A. longisetosus* was also found to synthesise oleic acid and LA (Brückner and Heethoff, 2020), but no ω3-PUFAs. In addition, the species *P. zaheri* and *M.* aff. *scutatus*, which as *Parasitus* sp. belong to the Mesostigmata (Acari), appear to be unable to synthesise PUFAs, especially ω3-PUFAs, *de novo* (Rueda-Ramírez et al., 2023a). Conversely, the present results provide first evidence for the synthesis of ω3-PUFAs in a mite, *Parasitus* sp.

In contrast to the studies discussed above, which were analysing fatty acid patterns in adult mites, this study focused on one of the immature stages of *Parasitus* sp., the deutonymph. The observed pattern points to alterations in fatty acid synthesis during ontogenetic development. Changes in fatty acid ratios during ontogenetic development have been observed in some insects, such as *Rhynchophus ferrugineus* Olivier (Curculionidae: Coleoptera) (Chen et al., 2024). Nevertheless, *Parasitus* sp. can synthesise the ω3-PUFAs ALA and ETA when fed with free-living nematodes, suggesting for the first time the presence of ω3 desaturases in mites. Yet, further studies are needed to confirm whether this species has this metabolic capacity throughout the entire life cycle or only during certain developmental stages.

4.4. Nematode diet rich in ω3 or ω6 LC-PUFA does not necessarily enhance Parasitus sp. fitness

The nematode diet appears to have a significant impact on the biological parameters of *Parasitus* sp. Fatty acid content plays a crucial role, but additional factors like other biochemical constituents of the nematodes or the nematodes' behaviour and size also influence these parameters. This is evidenced by the only partial correlation between the results of the life cycle experiments and the fatty acid pattern. For example, mites fed on *P. pacificus*, which has the highest ω3- PUFA, exhibited low survival. This discrepancy might be linked to the nematode's behavior, as it was observed to climb onto the mites in large numbers, potentially hindering *Parasitus* sp. development (Rueda-Ramírez et al., 2023a). A similar detrimental effect on development was noted in the coleopteran *R. ferrugineus*, which showed reduced larval lifespan when interacting with the nematode *Teratorhabditis synpapillata* (Sudhaus, 1985) (Rhabditida: Rhabditidae), which was previously considered phoretic (Ibrahim et al., 2021).

Nematode species, and their fatty acid content and profile, affected the life stages of *Parasitus* sp. in different ways (see Supp. Material S3.2). Among all nematode diets, *Panagrolaimus* sp. 1 was the most nutritionally beneficial for *Parasitus* sp., allowing rapid and consistent development into adulthood. The efficacy of this diet is probably due to its high ω3 LC-PUFA content, second only to *P. pacificus,* and also to its high total content of fatty acids (TLFA). The presence of LA in *Panagrolaimus* sp. 1, a precursor of PUFAs, may also have contributed to the observed rapid maturation. Notably, *de novo* synthesis of D-GLA occurred only in mites fed on this diet, along with relatively high levels of EPA, which likely supported oviposition. In insects, D-GLA, EPA, and ARA are precursors of hormones that stimulate egg-laying (Stanley and Kim, 2019b), and a similar mechanism may apply to *Parasitus* sp. In previous studies, *Panagrolaimus* sp. 1 was also a suitable diet for both *P. zaheri* and *M.* aff. *scutatus* (Rueda-Ramírez et al., 2023a).

The presence of some important PUFAs such as GLA in mites fed on *O. tipulae* appears to be associated with the observed low larval mortality and short protonymph stage. GLA is a precursor to tissue hormones involved in growth and immune function (Stanley and Kim, 2019b), which may explain this observations. However, D-GLA and ARA were absent in mites fed on *O. tipulae*, despite being present in the nematode. Since these fatty acids are important precursors to oviposition-stimulating hormones in insects (Stanley and Kim, 2019b), their absence might explain the lower fecundity compared to mites fed on *Panagrolaimus* sp. 1.

Although some biological parameters and fatty acid profiles of mites fed on *Panagrolaimus* sp. 2 were similar to those observed with *Panagrolaimus* sp. 1, the diet based on *Panagrolaimus* sp. 2 resulted in no oviposition and the shortest adult longevity of all treatments. This is likely attributed to the higher proportion of ω6-PUFAs relative to ω3-PUFAs in nematodes as well as mites compared to the *Panagrolaimus* sp. 1 diet. Mites fed on *Panagrolaimus* sp. 2 had high levels of GLA, which may explain their rapid development to adulthood. However, the low levels of EPA and ALA probably contributed to the short adult lifespan, while the low EPA content may also explain the lack of oviposition. Overall, the short lifespan appears to be the main limiting factor for reproduction.

De novo synthesis of fatty acids, especially ω3 LC-PUFAs, is metabolically demanding and entails some physiological cost (Reeds et al., 1982). This may explain the long duration and high mortality at the immature stages of mites fed diets lacking these essential molecules. Such a scenario appears to be the case for mites fed on *A. bodenheimeri*, which supported rapid and adequate development to the deutonymphal stage but was insufficient for most mites to undergo a third moult. No dietary ω3 LC-PUFAs combined with the nutritional and energetic demands associated with growth and moulting (Gillette et al., 2021;

Llandres et al., 2015; Lorenz and Gäde, 2009; Yuan et al., 2022) was likely insufficient to sustain *Parasitus* sp. development beyond the deutonymphal stage. Conversely, Rueda-Ramirez et al. (2023a) observed that this nematode was the diet with the highest quality for the development and fitness of *P. zaheri* and *M.* aff. *scutatus* populations. This strongly indicates that predatory mite species have different dietary requirements, which additionally can vary within the life cycle.

The prolonged duration of the immature stages of mites feeding *M. incognita*, with nearly all failing to reach adulthood, is likely due to the nematode's fatty acid profile, which was rich in MUFAs but deficient in ω6- and ω3-PUFAs, particularly GLA, LA, and ALA. The low GLA content may have contributed to the prolonged immature period, while the overall deficiency in PUFAs likely hindered mite development. Similar findings were reported for *M.* aff. *scutatus*, where *M. incognita* proved to be a low-quality food source, resulting in suboptimal life history parameters. However, as seen in other Macrochelidae species, members of Parasitidae could benefit from a combined diet, especially one including free-living nematodes, which significantly enhanced mite performance, e.g. favouring pest biocontrol in agroecosystems (Azevedo et al., 2019; Ito, 1973b; Rueda-Ramírez et al., 2023b).

4.5. Other factors may also be related to the survival and duration of immature stages of Parasitus sp

Although most of the nematode diets ensured post-embryonic development of mites, survival at some life stages as well as moulting to the adult stage was generally very low. This was evident in the survival curves, where populations such as the mites fed on *M. incognita* exhibited high mortality during immature stages, and some mites exhibited prolonged duration of certain life stages. A prolonged immature stage (up to six months), particularly the deutonymph phase, and low moulting success have been documented in other Parasitidae species when a certain stimulus (e.g. grouping individuals of different sexes, a change in diet, or simulation of phoresy) is absent prior to moulting to adulthood, or when resources or conditions are not optimal (Ito, 1973a, 1976, 1977a; Koulianos and Schwarz, 1999; Rueda-Ramírez et al., 2019; Walter and Proctor, 2013; Yasui, 2023). For example, as noted in Rueda-Ramírez et al. (2019), *P. bituberosus* has the fastest development, shortest pre-oviposition period, and longest oviposition period, when paired from the egg stage, instead of the deutonymph stage, as in the present study. Here, the low proportion of adults as well as the male-biased ratio of *Parasitus* sp. are similar to results observed in *P. bituberosus* when mites were not paired. It is therefore reasonable to hypothesize that *Parasitus* sp. would also positively benefit from being paired from the egg stage, resulting in improved life table parameters, mortality, as well as overall survival and fitness. However, cannibalism was observed in mites paired in younger stages, despite being fed *ad libitum,* therefore not allowing for such rearing conditions.

Finally, the limited diet, i.e. single nematode species, offered to *Parasitus* sp. may have contributed to the low maturation rates observed. Higher fecundity in *P. bituberosus* and thrips *Frankliniella occidentalis* (Pergande, 1895) (Insecta: Thysanoptera: Thripidae) has been reported when fed a mixed diet of a free-living nematode (Rueda-Ramírez et al., 2019). The latter is supported by the successful growth and maintenance of colonies in the laboratory fed on a mixture of free-living nematode species. Thus, a varied diet, especially including nematodes in their immature stages, is important for mite fitness (Ito, 1977b; Rueda-Ramírez et al., 2023b).

The diet of the nematode prey also plays a role in the fitness of their mite predator. The nematodes on which the mite colonies were maintained were from mass cultures on a non-sterile medium consisting of decomposing oat flakes, with an unidentified diverse microbial community. In contrast, to ensure a homogenous fatty acid pattern, the nematodes used in the experiments were obtained from monoxenic cultures on *Escherichia coli* Migula, 1895 Castellani and Chalmers 1919 (Bacteria: Enterobacteriaceae) strain OP50. Azevedo et al. (2019) observed that the fecundity of *Macrocheles embersoni* Azevedo, Berto and Castilho, 2018 (Acari: Mesostigmata: Macrochelidae) was higher when their nematode prey *R. axei* was reared on decomposing beans with a varied bacterial community, compared to being reared on yeast alone. The better nutritional quality of a diverse bacterial community for their nematode grazers may cascade up the food chain with positive implications for higher trophic levels. The effect of diversity and quality at lower trophic levels on higher trophic levels has been observed in studies such as that of Lu et al. (2022), in which variations in plant quality influenced herbivore taxon richness, in turn influencing predator taxon richness. This suggests that nutrient availability at lower trophic levels can have significant effects on the entire food web.

5. Conclusions

This study links the effects of diet quality with the fitness of predatory mites in soil food webs. In line with previous findings, diet quality varied depending on nematode species, emphasizing the importance of nematode diversity in soil ecosystems for the performance, development, and survival of their predators. While some of the same nematode diets supported both development and optimal adult stages in other mites previously evaluated, *Parasitus* sp. only reached certain stages before survival declined, suggesting that a nutritious food source is not universally beneficial for all predators. Apart from that, *Parasitus* sp. exhibited the best development and oviposition when feeding on nematodes rich in PUFAs, particularly ω3 LC-PUFAs, confirming the hypothesis of the importance of these fatty acids in supporting this mite fitness. In contrast, nematodes lacking ω3-PUFAs but containing high levels of ω6-PUFAs were less suitable for mite fitness, possibly due to the physiological stress of synthesizing these nutritionally valuable fatty acids. Furthermore, nematode prey low in both ω3 and ω6 LC-PUFAs had the poorest diet quality.

Our study provides novel insights into the lipid metabolism of *Parasitus* sp. deutonymphs and demonstrates a strong trophic link between the fatty acid composition of this developmental stage and that of its nematode prey. Notably, this is the first study to demonstrate ω3-PUFA synthesis in a Mesostigmata mite and, due to that, our findings provide partial support for the hypothesis that this predatory mites reflect the ω3 LC-PUFA content of the diet. However, further research is needed to clarify how nutritional requirements change during mite ontogeny and how mixed diets affect their life cycle.

Funding

This work was supported by German Research Foundation (DFG) [grant number RU780/23–1] ('Quantitative fatty acid signature analysis: a new tool to determine tropic interactions in soil food webs').

CRediT authorship contribution statement

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests

Diana Rueda-Ramirez reports financial support was provided by

German Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

To Rainer Nehring, for his outstanding technical assistance in all GC/MS experiments. To Ralph Sommer, Dept. for Integrative Evolutionary Biology Max-Planck Institute for Biology Tübingen, Germany, for his remarkable assistance in the identification of *Pristionchus pacificus*. To Johannes Hallmann for his inestimable help with the plant-parasitic nematodes and valuable suggestions during the work and to the manuscript. This work was supported by German Research Foundation (DFG) [grant number RU780/23–1] ('Quantitative fatty acid signature analysis: a new tool to determine tropic interactions in soil food webs').

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.pedobi.2025.151038.

Data availability

Data will be made available on request.

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